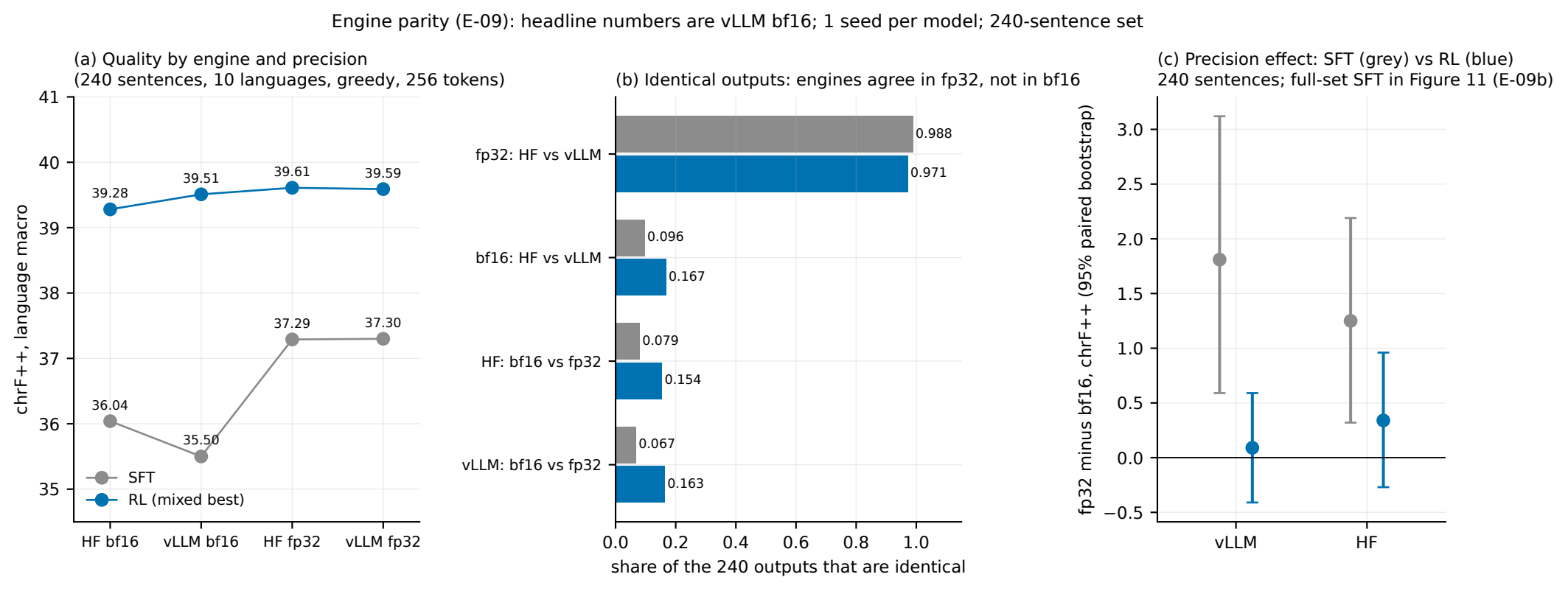
Engine parity (E-09): headline numbers are vLLM bf16; 1 seed per model; 240-sentence set
(a) Quality by engine and precision
(240 sentences, 10 languages, greedy, 256 tokens)
chrF++, language macro
41
40
39
38
37
36
35
39.28
39.51
39.61
39.59
37.29
37.30
36.04
35.50
SFT
RL (mixed best)
HF bf16
vLLM bf16
HF fp32
vLLM fp32
(b) Identical outputs: engines agree in fp32, not in bf16
fp32: HF vs vLLM
0.988
0.971
bf16: HF vs vLLM
0.096
0.167
HF: bf16 vs fp32
0.079
0.154
vLLM: bf16 vs fp32
0.067
0.163
0.0
0.2
0.4
0.6
0.8
1.0
share of the 240 outputs that are identical
(c) Precision effect: SFT (grey) vs RL (blue)
240 sentences; full-set SFT in Figure 11 (E-09b)
fp32 minus bf16, chrF++ (95% paired bootstrap)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
vLLM
HF